

Brill–Noether Theory for chains of loops

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Recall from last time if you were listening/care

A divisor D on a curve X is a formal sum

$$D = \sum_{p \in X} a_p p.$$

The degree of D is the sum over all coefficients.

For a nonzero rational function, f the divisor of f is given by

$$\operatorname{div}(f) = \sum_{p \in X} \operatorname{ord}_p(f) p.$$

Finally, the point of study for these curves is the space of functions

$$L(D) = \{f \text{ meromorphic on } X : \operatorname{div}(f) + D \geq 0\}.$$

We define $\ell(D) = \dim L(D)$, and $r(D) = \ell(D) - 1$.

Classical motivation

Let X be a smooth projective curve of genus g . Define degree- d Picard variety

$$\mathrm{Pic}_d(X) = \{[D] \in \mathrm{Div}(X)/\sim \mid \deg(D) = d\}.$$

We want to study the geometry of the *Brill–Noether locus*,

$$W_d^r = \{[D] \in \mathrm{Pic}_d(X) \mid r(D) \geq r\}$$

The elements of which, describe maps

$$X \rightarrow \mathbb{P}^r$$

The Brill–Noether number,

$$\rho = g - (r + 1)(g - d + r)$$

roughly estimates the dimension of the Brill–Noether locus $W_d^r(X)$.

Classical motivation (cont.)

For algebraic curves, the Brill–Noether number

$$\rho = g - (r + 1)(g - d + r)$$

predicts the geometry of $W_d^r(X)$.

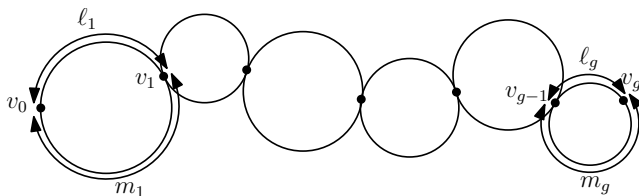
Theorem (Griffiths–Harris 1980)

Let X be a general curve

- (1) If $\rho < 0$, then $W_d^r = \emptyset$.
- (2) Else, $\dim W_d^r = \min\{\rho, g\}$.

Now, degenerate to a tropical curve. Does such a theorem still hold true?

A Tropical Brill–Noether Theorem



A *generic* chain of loops behaves as a generic curve X does.

Theorem (Cools–Draisma–Payne–Robeva 2012)

Let Γ be a generic chain of loops of genus g .

- (1) If $\rho < 0$, then $W_d^r(\Gamma) = \emptyset$.
- (2) Else, $\dim W_d^r(\Gamma) = \min\{\rho, g\}$.

Example: $(g, r, d) = (4, 1, 3)$

Fact

When $\rho = 0$, these maps are in bijection with standard Young tableaux of shape

$$(r + 1) \times (g - d + r).$$

Notice, $\rho = 4 - 2(4 - 3 + 1) = 0$. Thus the relevant tableaux have shape

$$(r + 1) \times (g - d + r) = 2 \times 2.$$

Recall that a standard young tableaux must be filled with alphabet $(1, \dots, g)$ where the columns and rows are increasing. We only have

1	2
3	4

1	3
2	4

By the tropical BN theorem, these are the two divisor classes in $W_3^1(\Gamma)$.

The classical picture

Let C_4 be a general curve of genus 4. We have a canonical embedding

$$C_4 \hookrightarrow \mathbb{P}^3.$$

Moreover, it is known that C_4 lies on a smooth quadric

$$Q \cong \mathbb{P}^1 \times \mathbb{P}^1.$$

This quadric has two distinct rulings, which induce a g_3^1 on C_4 . It is true, but nontrivial to show, that these correspond exactly to our tableaux!

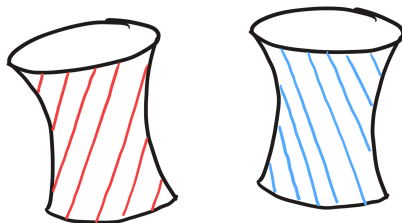
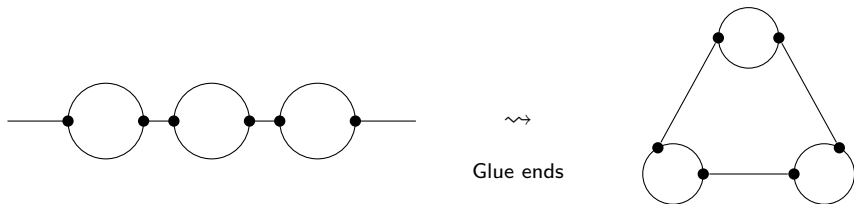


Fig: The two rulings of $\mathbb{P}^1 \times \mathbb{P}^1$

Next steps

As the motivating object behind tropical Brill–Noether theory, divisors on chains of loops are very well studied. So are chains of loops with bridges.

Consider a chain of loops with bridges of genus g . Then observe the “cyclic chain of loops” obtained by gluing the two end bridges together.



Question

What can Brill–Noether theory say about these graphs? As far as we know it hasn't been studied.

References



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F. Cools, J. Draisma, S. Payne, and E. Robeva.

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Thank You!